

TurtleBot3 Gazebo SLAM & Navigation



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TurtleBot3 Gazebo SLAM & Navigation

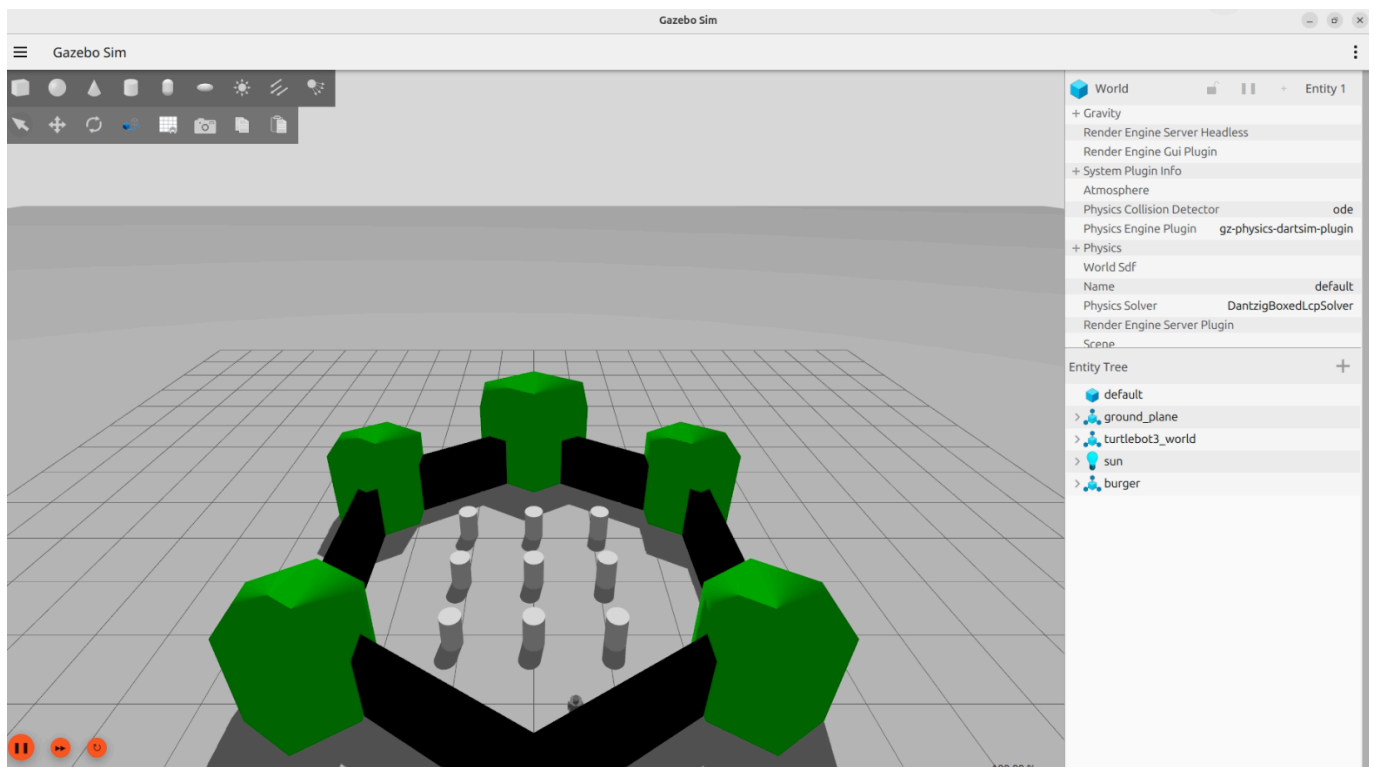
Exported from Confluence · <https://pinkwink.atlassian.net>

TurtleBot3 SLAM simulation

거북이 세상 가제보 실행

Code

```
1 ros2 launch turtlebot3_gazebo turtlebot3_world.launch.py
```



그런데 혹시나 가제보에서 이전에 작업하던 환경이 불러와진다면..

gz 이름이 들어간 프로세스들을 모두 말살하고 다시 시작해야한다.

Code

```
1 sudo pkill -9 -f "gz|gazebo|ign"
```

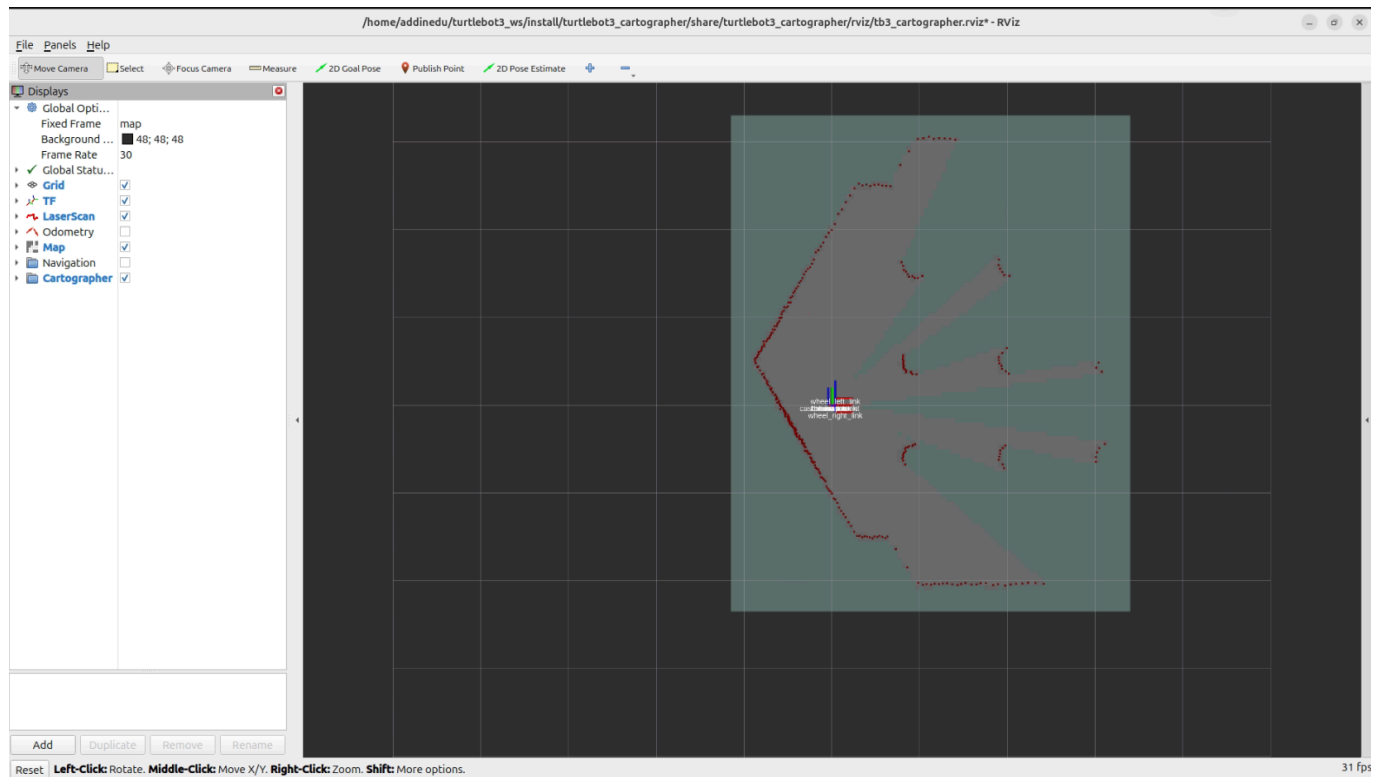


```
ham@pc:~$ ps aux | grep gz
addnedu 48877 0.6 1.8 4444972 283968 ? Sl Apr27 5:27 gz sim -r -s -v4, /home/addnedu/pinky/install/pinky_gz_sim/share/pinky_gz_sim/worlds/pinky_factory.world
addnedu 49689 1.7 1.1 4444972 177592 ? Sl Apr27 13:42 gz sim -r -s -v4, /home/addnedu/pinky/install/pinky_gz_sim/share/pinky_gz_sim/worlds/pinky_factory.world
addnedu 51308 136 2.2 4883564 350056 ? Sl Apr27 1082:36 gz sim -r -s -v4, /home/addnedu/pinky/install/pinky_gz_sim/share/pinky_gz_sim/worlds/pinky_factory.world
addnedu 53629 9.3 29.6 11160072 4611992 ? Sl Apr27 72:51 gz sim -g -v4
addnedu 54872 0.0 0.4 4199196 76412 ? Sl Apr27 0:42 gz sim -r -s -v4, /home/addnedu/pinky/install/pinky_gz_sim/share/pinky_gz_sim/worlds/pinky_factory.world
addnedu 54873 0.1 0.7 6458240 113612 ? Sl Apr27 1:23 gz sim -g -v4
addnedu 64491 0.0 0.0 10800 2348 pts/0 S+ 04:44 0:00 grep --color=auto gz
ham@pc:~$ kill -9 48877 49689 51308 53629 54872 54873
```

터미널 하나 더 열어서 SLAM 실행

Code

```
1 ros2 launch turtlebot3_cartographer cartographer.launch.py use_sim_time:=True
```



터미널 하나 더 열어서 터틀봇 전용 teleop keyboard 실행

```
/bin/bash

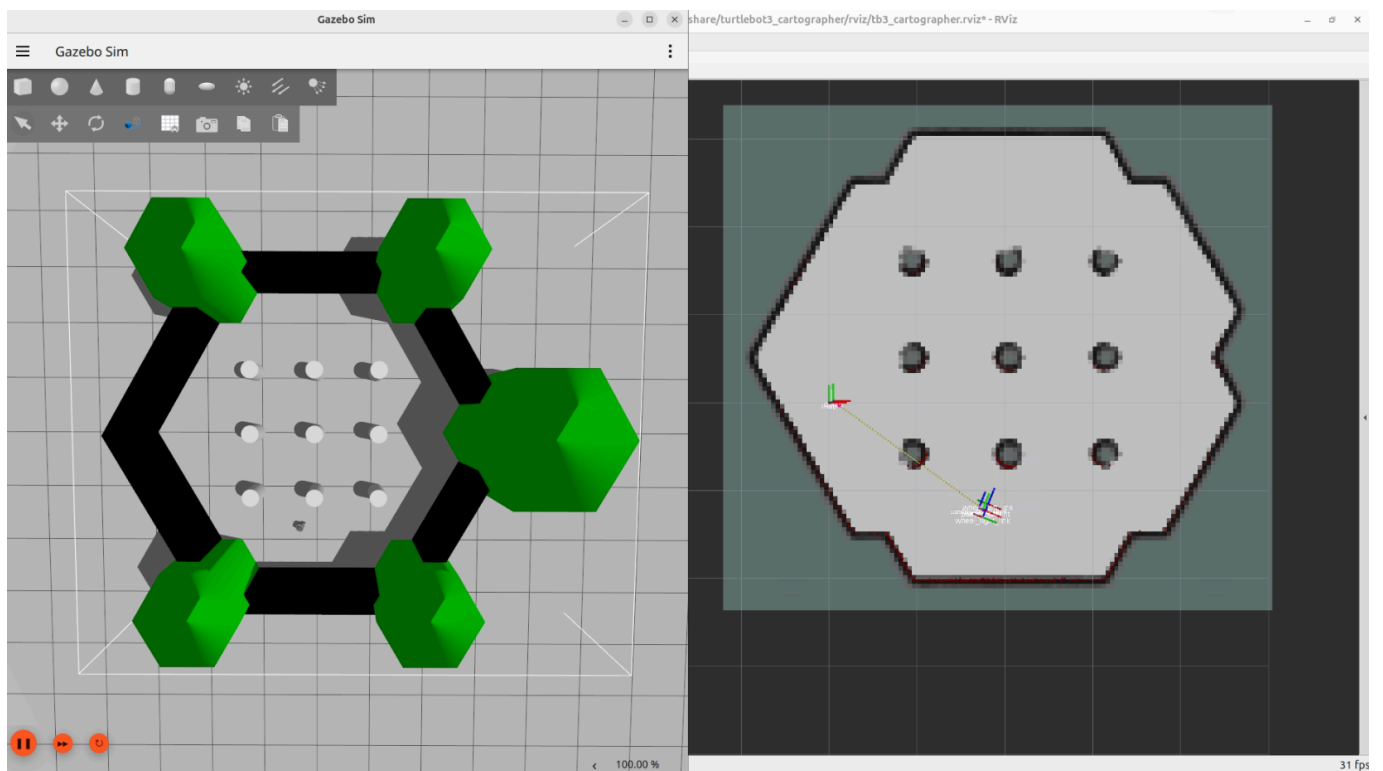
ham@pc:~$ ros2 run turtlebot3_teleop teleop_keyboard

Control Your TurtleBot3!
-----
Moving around:
   w      a      s      d      x
   |      |      |      |      |
w/x : increase/decrease linear velocity (Burger : ~ 0.22, Waffle and Waffle Pi : ~ 0.26)
a/d : increase/decrease angular velocity (Burger : ~ 2.84, Waffle and Waffle Pi : ~ 1.82)

space key, s : force stop

CTRL-C to quit
```

터틀봇3 키보드 제어 터미널에 w, a, s, d, x로 움직이면서 맵 생성하기



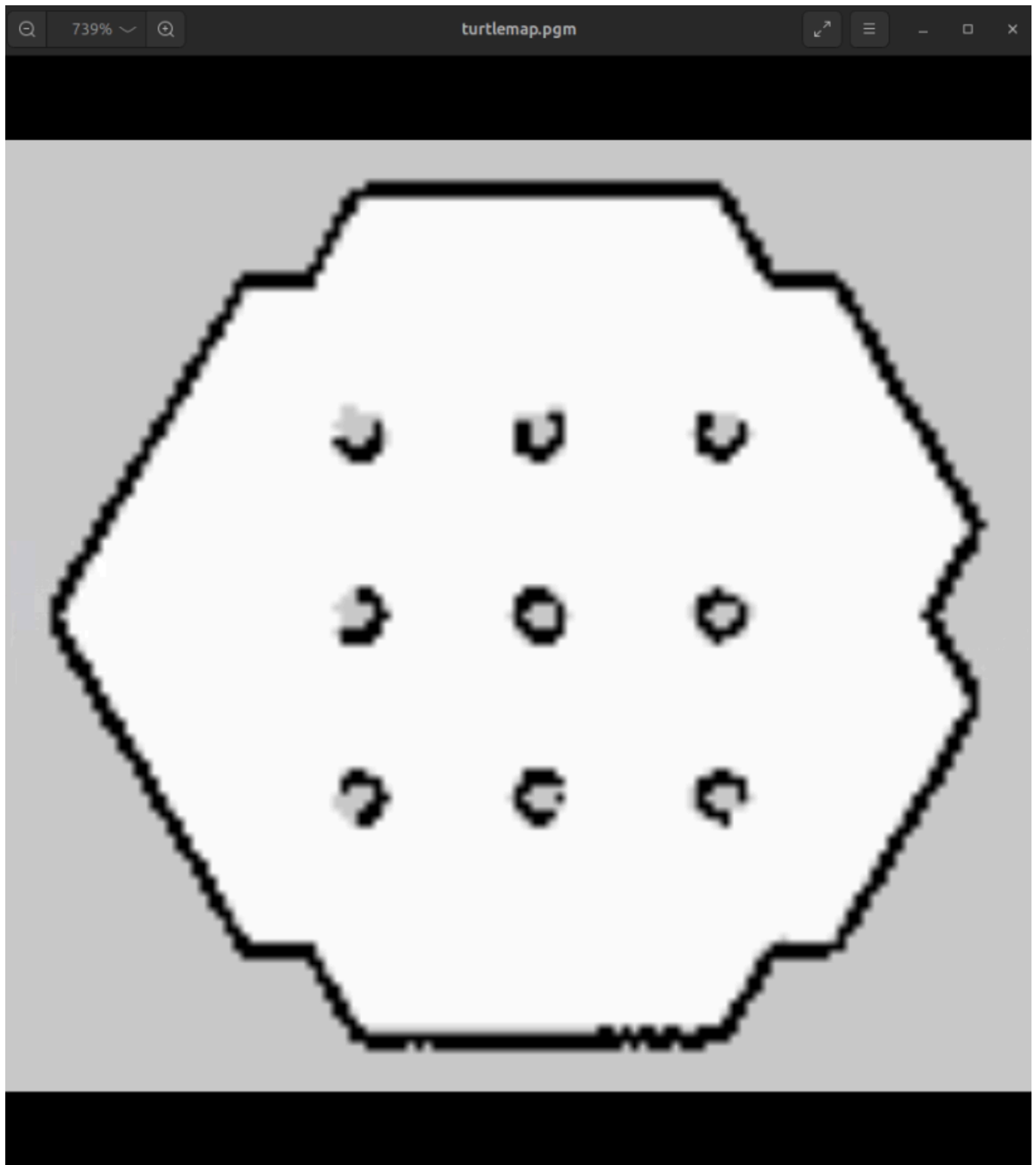
완성됐으면 맵 저장

Code

```
1 ros2 run nav2_map_server map_saver_cli -f ~/turtlemap
```

```
ham@pc:~$ ros2 run nav2_map_server map_saver_cli -f ~/turtlemap
[INFO] [1777320228.189338318] [map_saver]:
    map_saver lifecycle node launched.
    Waiting on external lifecycle transitions to activate
    See https://design.ros2.org/articles/node_lifecycle.html for more information.
[INFO] [1777320228.189484838] [map_saver]: Creating
[INFO] [1777320228.189538881] [map_saver]: Configuring
[INFO] [1777320228.197783977] [map_saver]: Saving map from 'map' topic to '/home/addinedu/turtlemap' file
[WARN] [1777320228.197823222] [map_saver]: Free threshold unspecified. Setting it to default value: 0.250000
[WARN] [1777320228.197836708] [map_saver]: Occupied threshold unspecified. Setting it to default value: 0.650000
[WARN] [1777320228.370063136] [map_io]: Image format unspecified. Setting it to: pgm
[INFO] [1777320228.370391773] [map_io]: Received a 125 X 115 map @ 0.05 m/pix
[INFO] [1777320228.394979771] [map_io]: Writing map occupancy data to /home/addinedu/turtlemap.pgm
[INFO] [1777320228.395484634] [map_io]: Writing map metadata to /home/addinedu/turtlemap.yaml
[INFO] [1777320228.395613179] [map_io]: Map saved
[INFO] [1777320228.395625643] [map_saver]: Map saved successfully
[INFO] [1777320228.397449281] [map_saver]: Destroying
```

꼭 저장된 거 잘 확인한 후에 실행한 터미널들 종료!



TurtleBot3 Navigation simulation

내비게이션 실행

- 저장한 맵 이름의 yaml 파일로 바꿔서 저장한 위치에서 실행

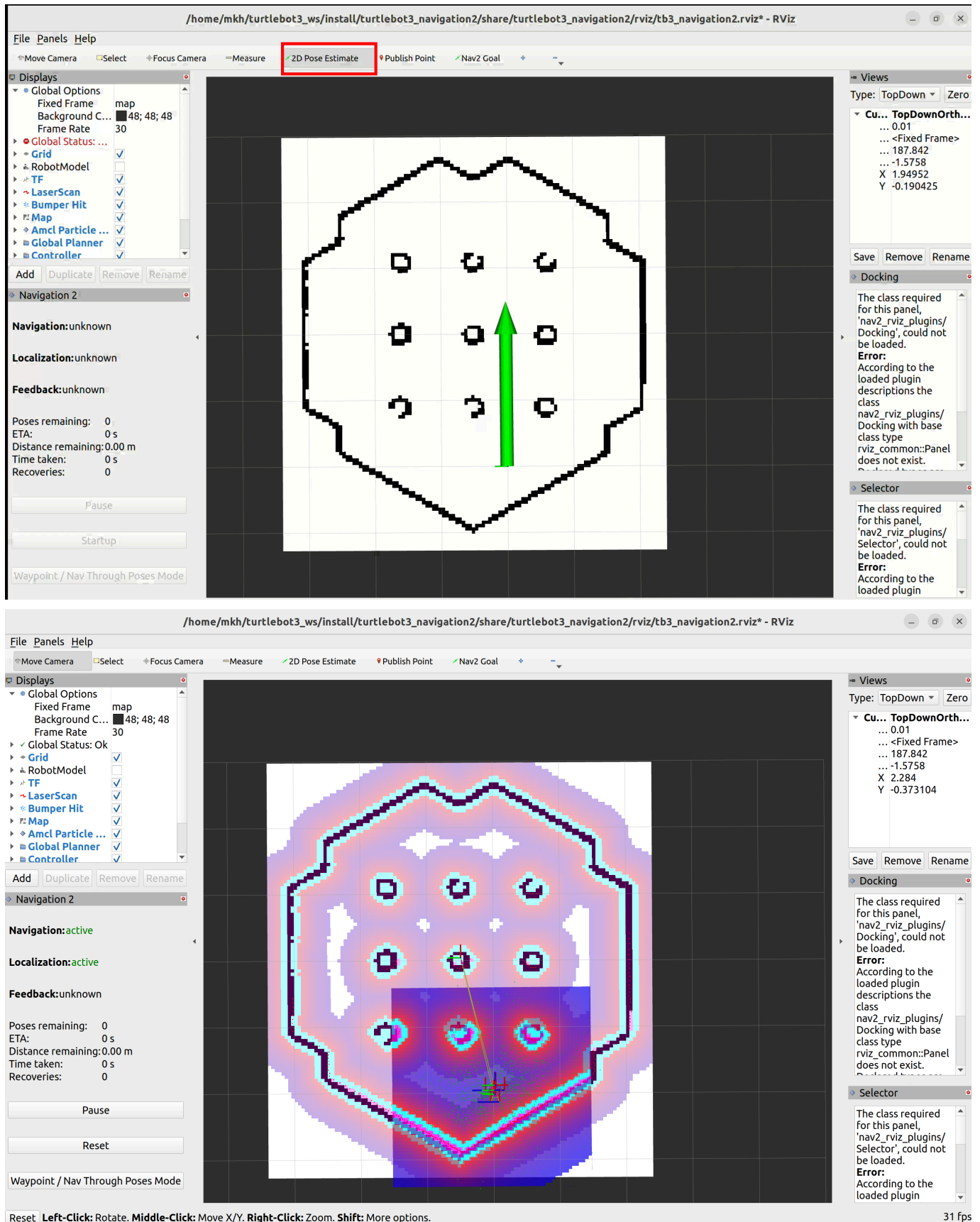
Code

```
1 ros2 launch turtlebot3_navigation2 navigation2.launch.py use_sim_time:=True map:=~/turtlemap.yaml
```

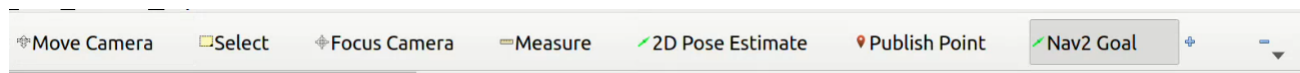
```
mkh@mkh:~/turtlebot3_ws$ ros2 launch turtlebot3_navigation2 navigation2.launch.py use_sim_time:=True map:=my_map.yaml
[INFO] [launch]: All log files can be found below /home/mkh/.ros/log/2025-07-14-16-04-03-345923-mkh-46568
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [component_container_isolated-1]: process started with pid [46582]
[INFO] [rviz2-2]: process started with pid [46584]
[rviz2-2] Warning: Ignoring XDG_SESSION_TYPE=wayland on Gnome. Use QT_QPA_PLATFORM=wayland to run on Wayland anyway.
[component_container_isolated-1] [INFO] [1752476643.892527430] [nav2_container]: Load Library: /opt/ros/humble/lib/libmap_server_core.so
[component_container_isolated-1] [INFO] [1752476643.900415386] [nav2_container]: Found class: rclcpp_components::NodeFactoryTemplate<nav2_map_server::CostmapFilterInfoServer>
[component_container_isolated-1] [INFO] [1752476643.900461886] [nav2_container]: Found class: rclcpp_components::NodeFactoryTemplate<nav2_map_server::MapSaver>
[component_container_isolated-1] [INFO] [1752476643.900466969] [nav2_container]: Found class: rclcpp_components::NodeFactoryTemplate<nav2_map_server::MapServer>
[component_container_isolated-1] [INFO] [1752476643.900470861] [nav2_container]: Instantiate class: rclcpp_components::NodeFactoryTemplate<nav2 map
```

The screenshot shows the RViz2 interface running in a terminal window. The main display area shows a 2D map with a black robot icon and a green path. The left sidebar contains a 'Displays' panel with a tree view of the loaded plugins, including 'Global Options', 'Grid', 'RobotModel', 'TF', 'LaserScan', 'Bumper Hit', 'Map', 'Amcl Particle Filter', 'Global Planner', and 'Controller'. Below this is a 'Navigation 2' panel showing status information: 'Navigation: unknown', 'Localization: unknown', 'Feedback: unknown', and 'Poses remaining: 0'. The right sidebar contains a 'Views' panel showing the current view type as 'TopDownOrtho' and a 'Docking' panel with error messages. The bottom status bar shows 'Reset', 'Left-Click: Rotate', 'Middle-Click: Move X/Y', 'Right-Click: Zoom', 'Shift: More options', and '31 fps'.

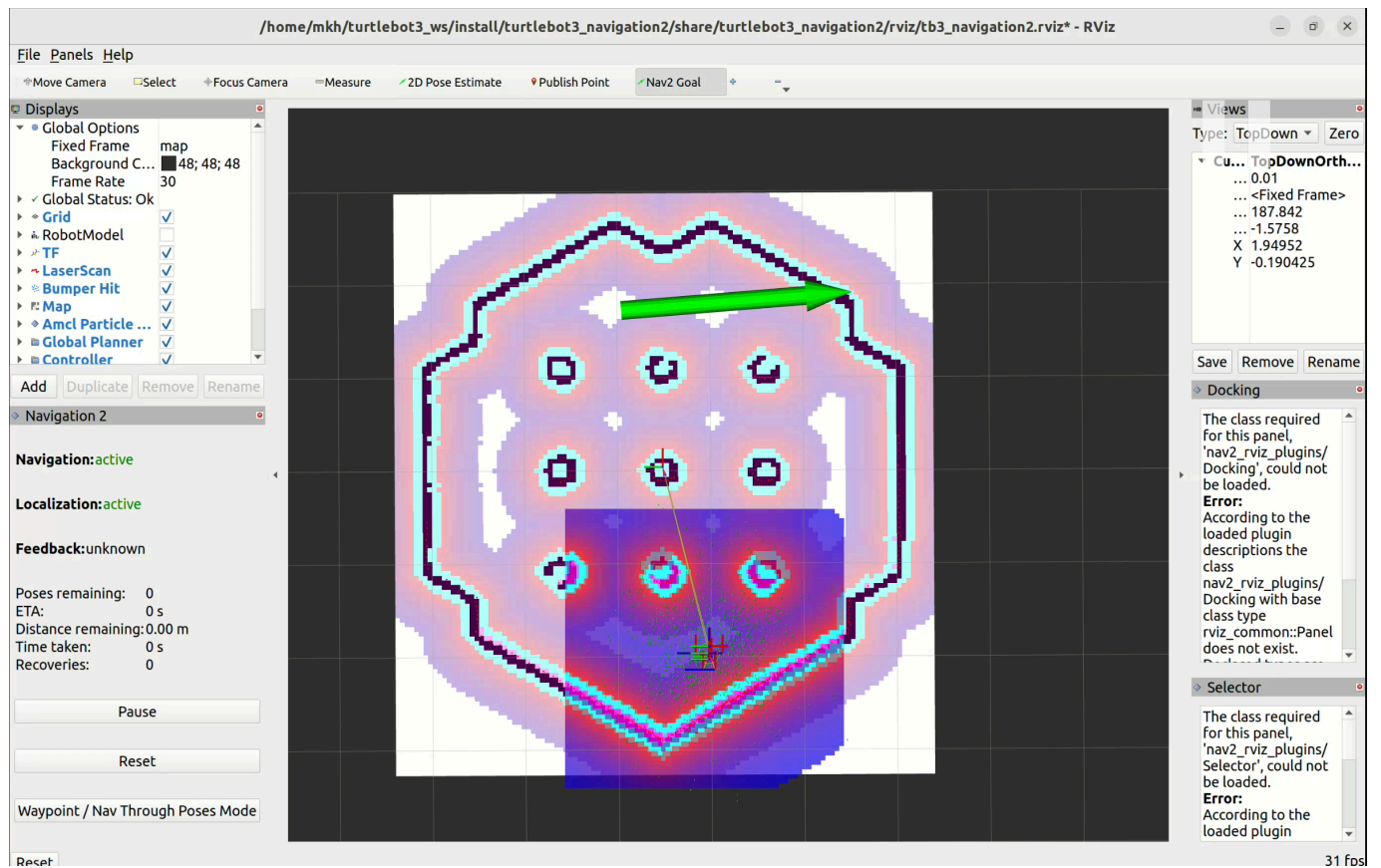
2D Pose Estimate 버튼을 눌러 맵위의 로봇의 위치와 방향을 설정하기



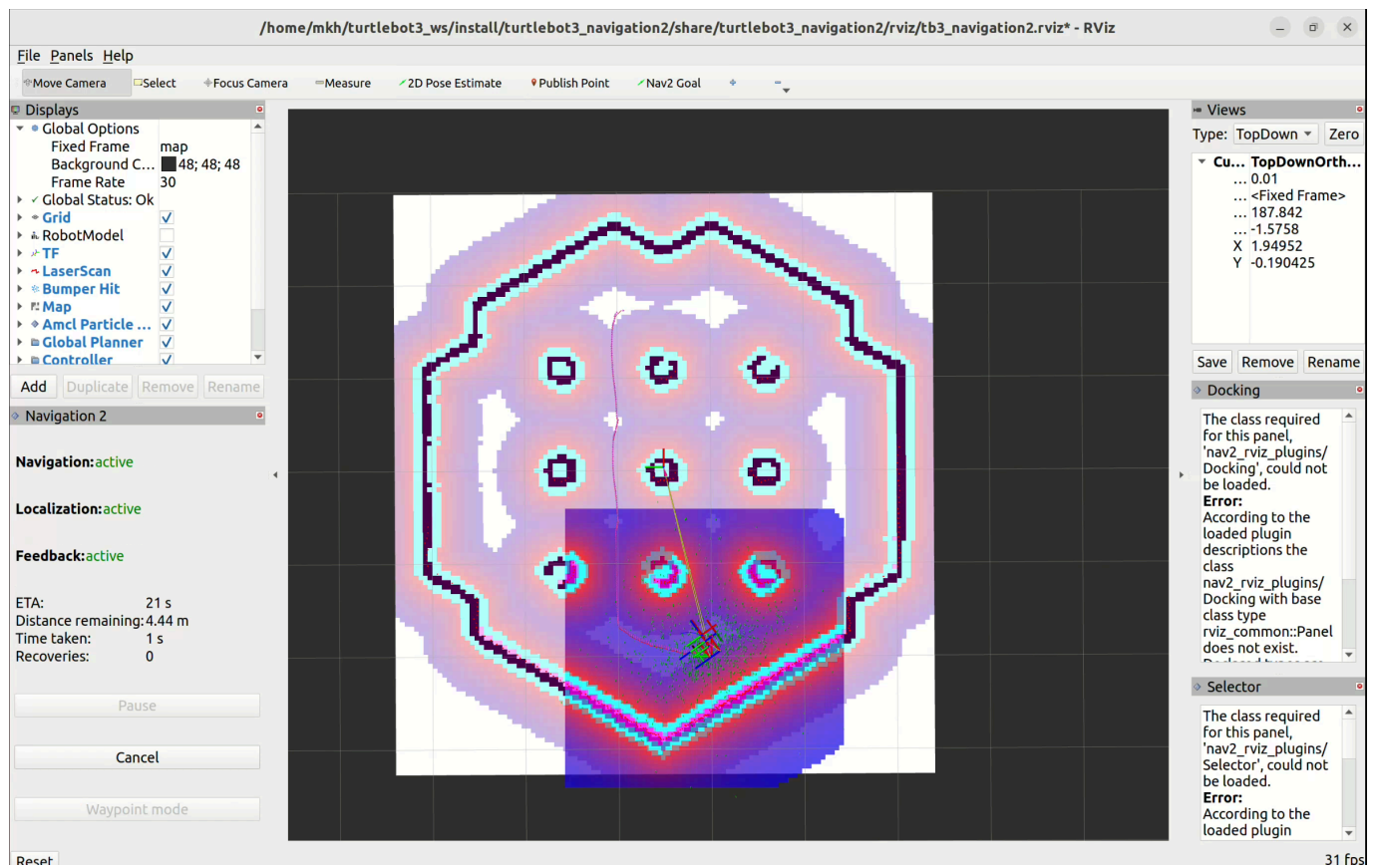
그 다음 Nav2 Goal를 클릭하고



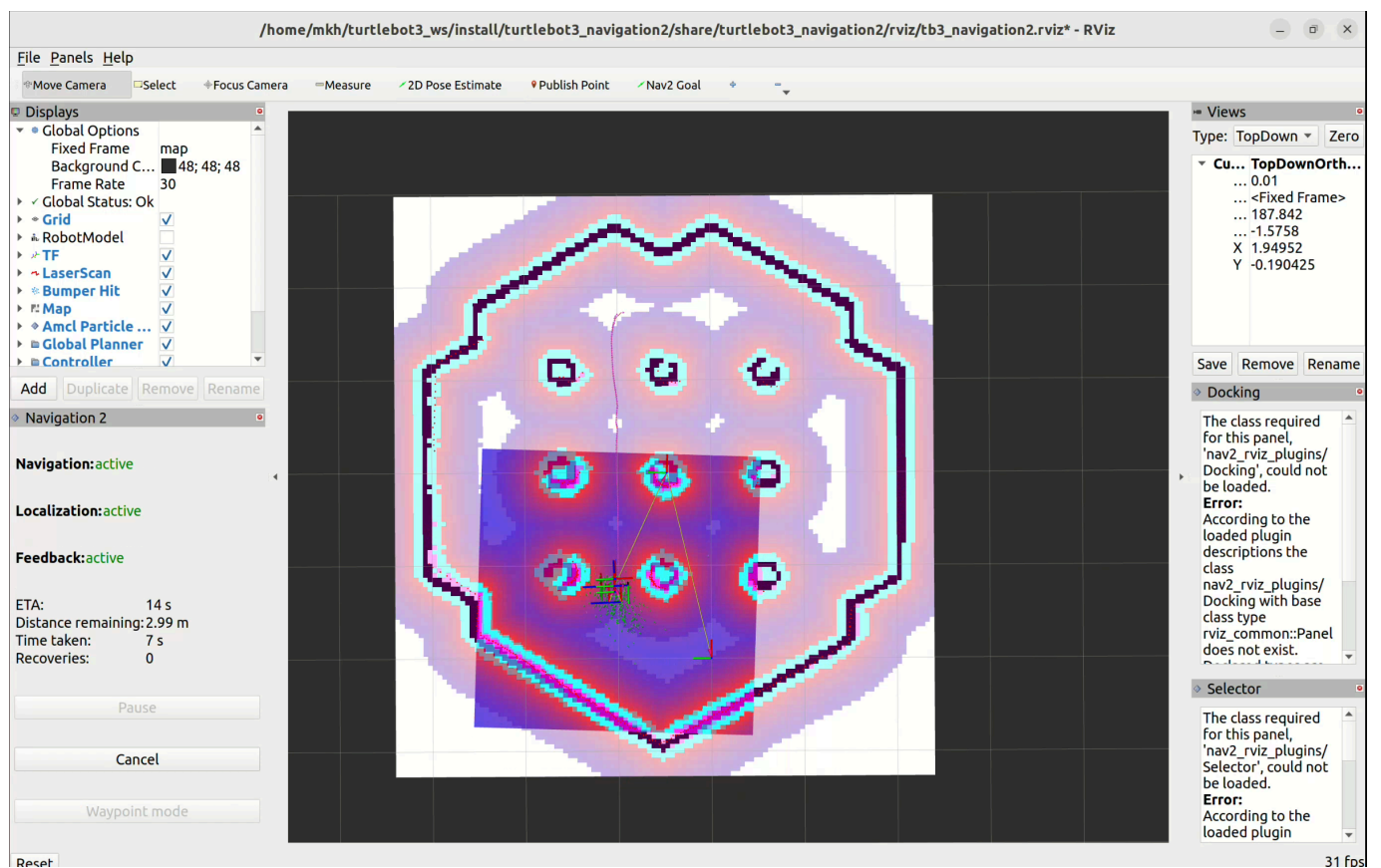
맵 위에 도착할 위치와 로봇의 방향을 설정하면



맵 위에 경로가 생기고



이동하기 시작한다



주행 중에 현재 로봇이 위치한 좌표를 알고싶으면 `amcl_pose` 토픽 구독하기

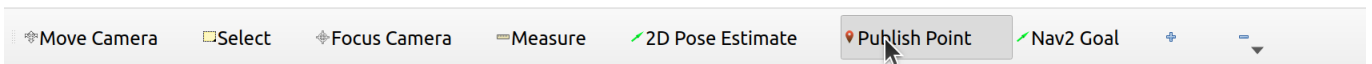
```
mkh@mkh:~/turtlebot3_ws$ ros2 topic echo /amcl_pose
header:
  stamp:
    sec: 1706785665
    nanosec: 813986619
  frame_id: map
pose:
  pose:
    position:
      x: -0.2992126467269522
      y: 1.525236013761885
      z: 0.0
    orientation:
      x: 0.0
      y: 0.0
      z: 0.09122559924502945
      w: 0.9958302516204182
  covariance:
    - 0.27267148404196645
    - 0.011542218527023418
    - 0.0
    - 0.0
```

맵 위에 어떤 지점의 좌표를 알고싶을 때는 `clicked_point` 라는 토픽을 구독하고

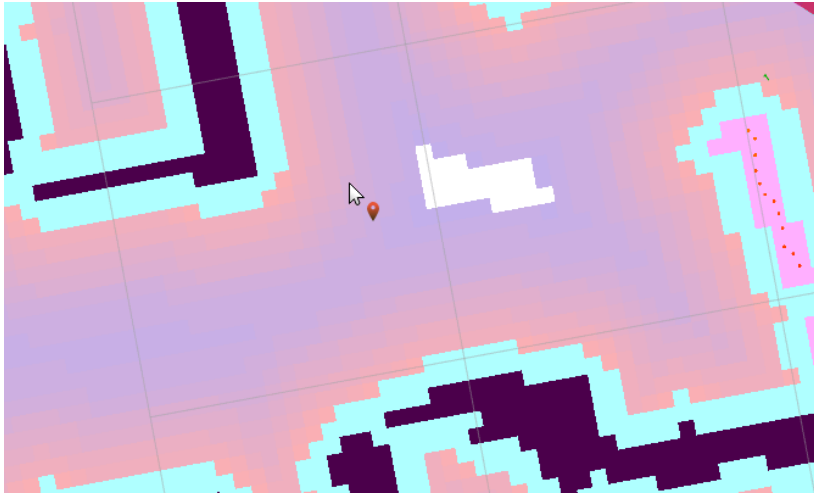
```
mkh@mkh:~/turtlebot3_ws$ ros2 topic echo /clicked_point
```

```
[ ]
```

publish point를 클릭하고



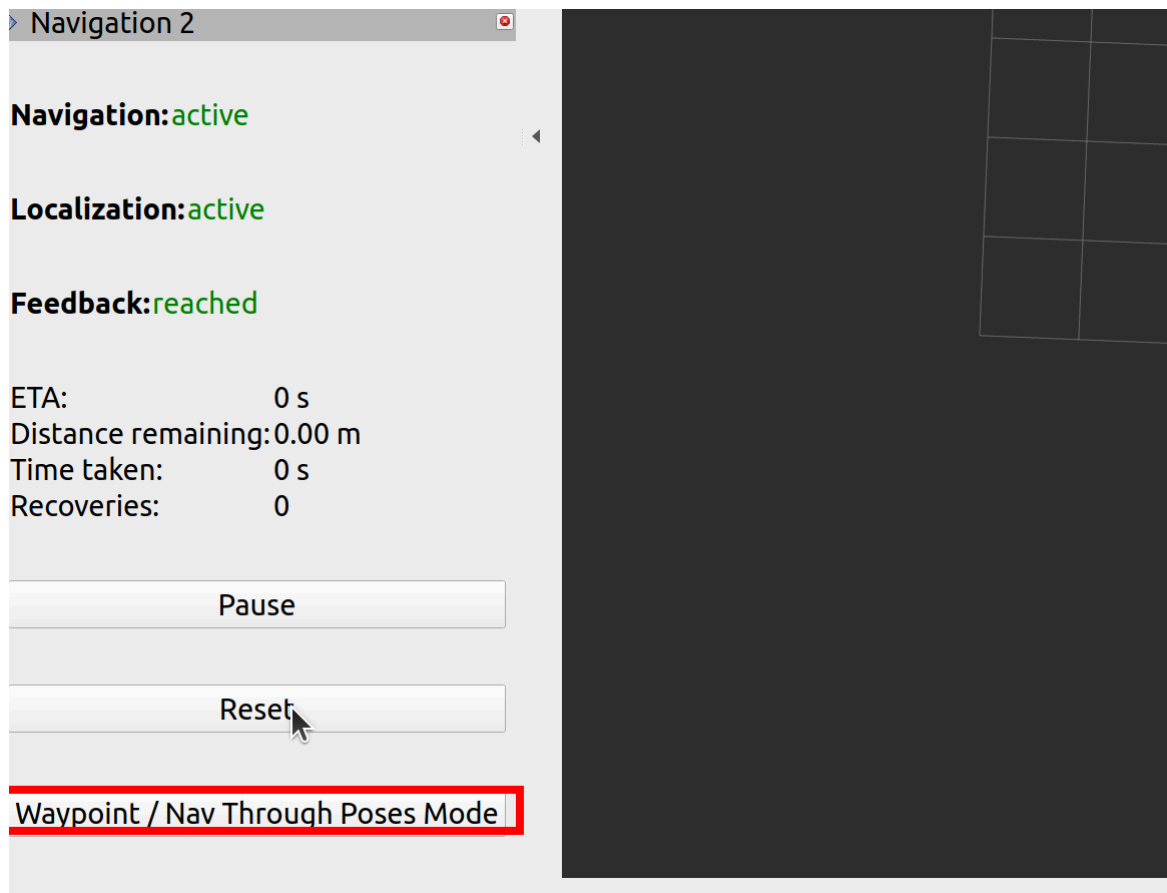
맵 위에 알고싶은 좌표를 클릭



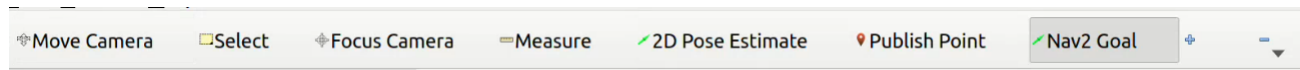
그리고 `clicked_point` 토픽 구독한 터미널을 보면 클릭한 좌표를 알 수 있다

```
mkh@mkh:~/turtlebot3_ws$ ros2 topic echo /clicked_point
header:
  stamp:
    sec: 1706784885
    nanosec: 2030787
  frame_id: map
point:
  x: -0.36737552285194397
  y: 2.600782632827759
  z: -0.001434326171875
---
```

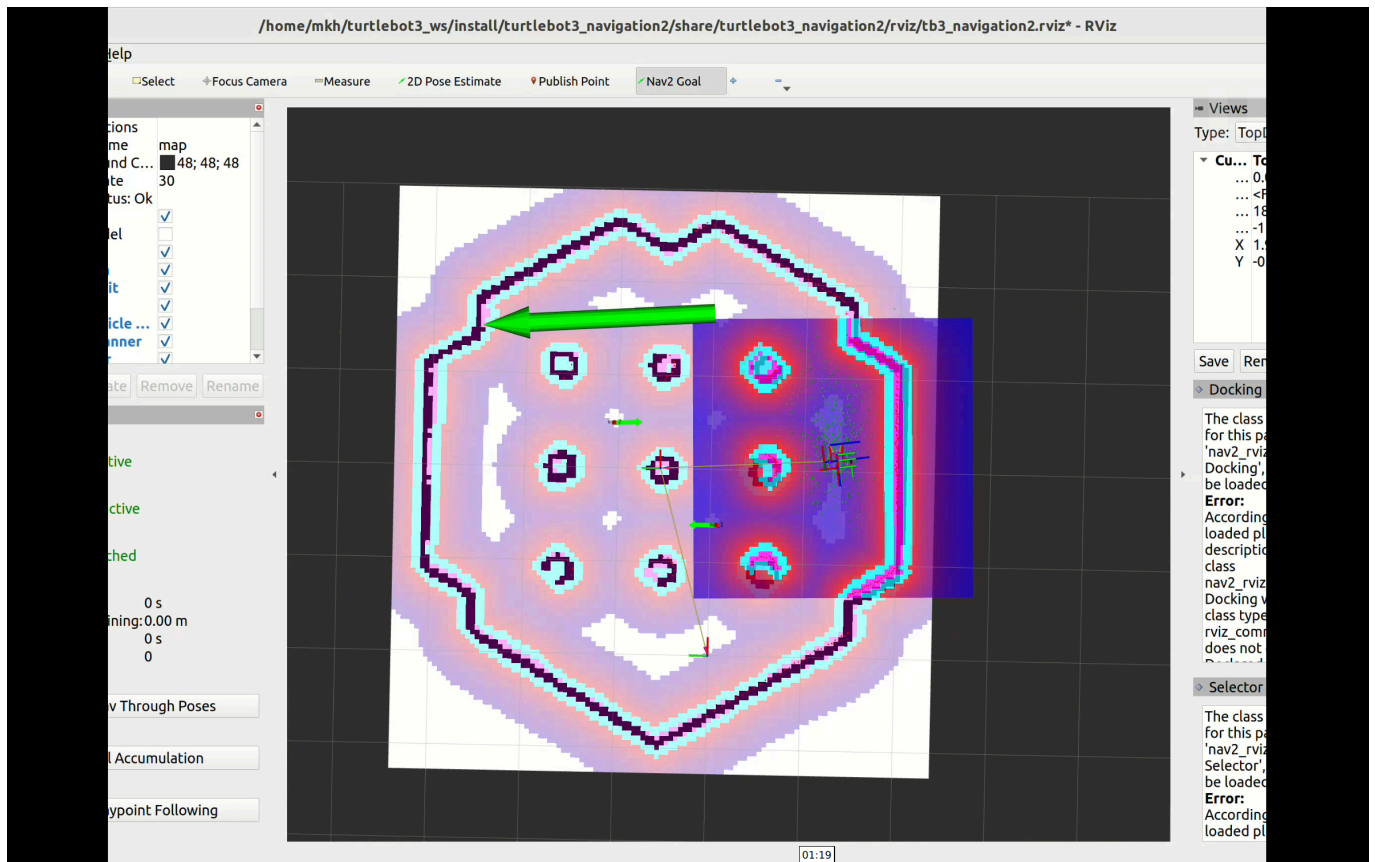
만약 한번에 여러 좌표로 보내고 싶으면 왼쪽의 `waypoint`를 클릭하고



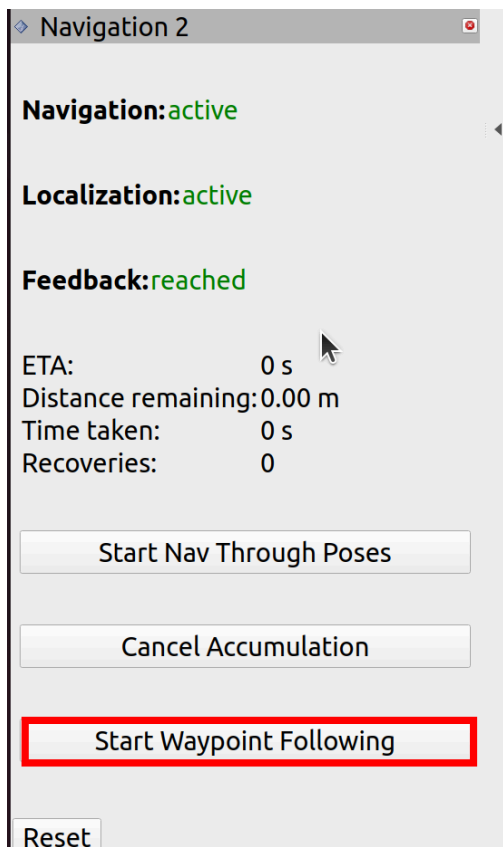
다시 Nav2 Goal를 클릭하고



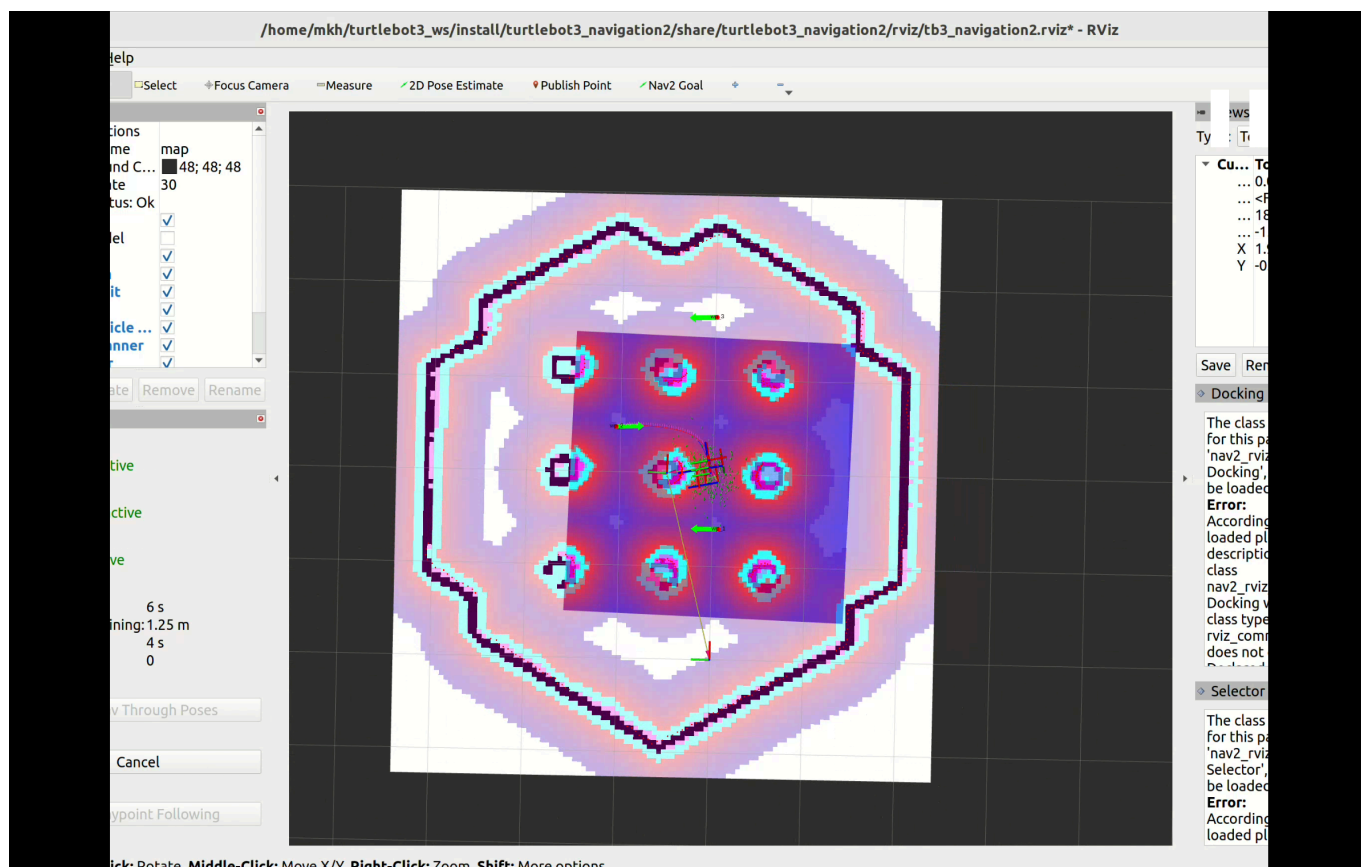
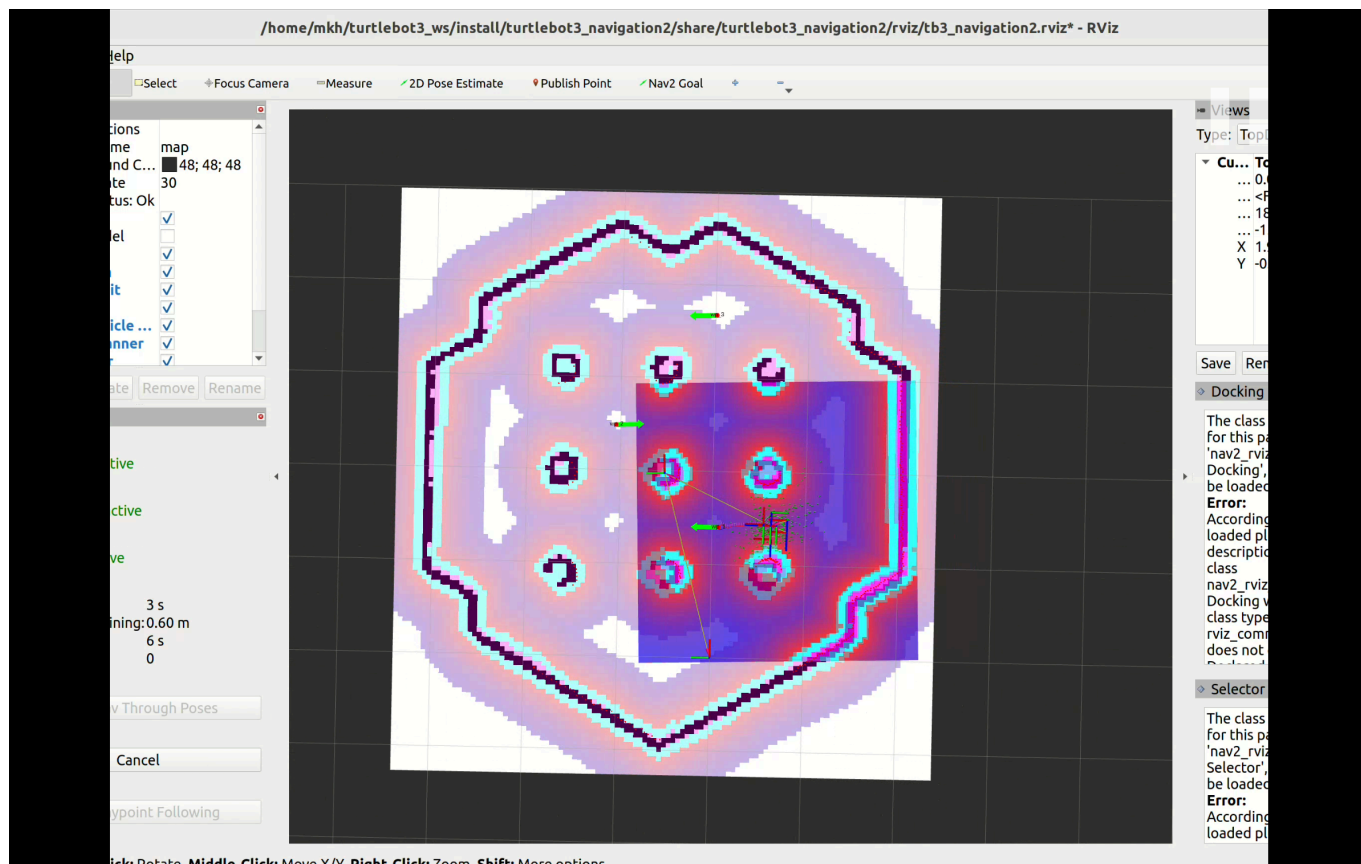
waypoint를 하나씩 추가하면 아래처럼 여러 개의 좌표를 지정 할 수 있다

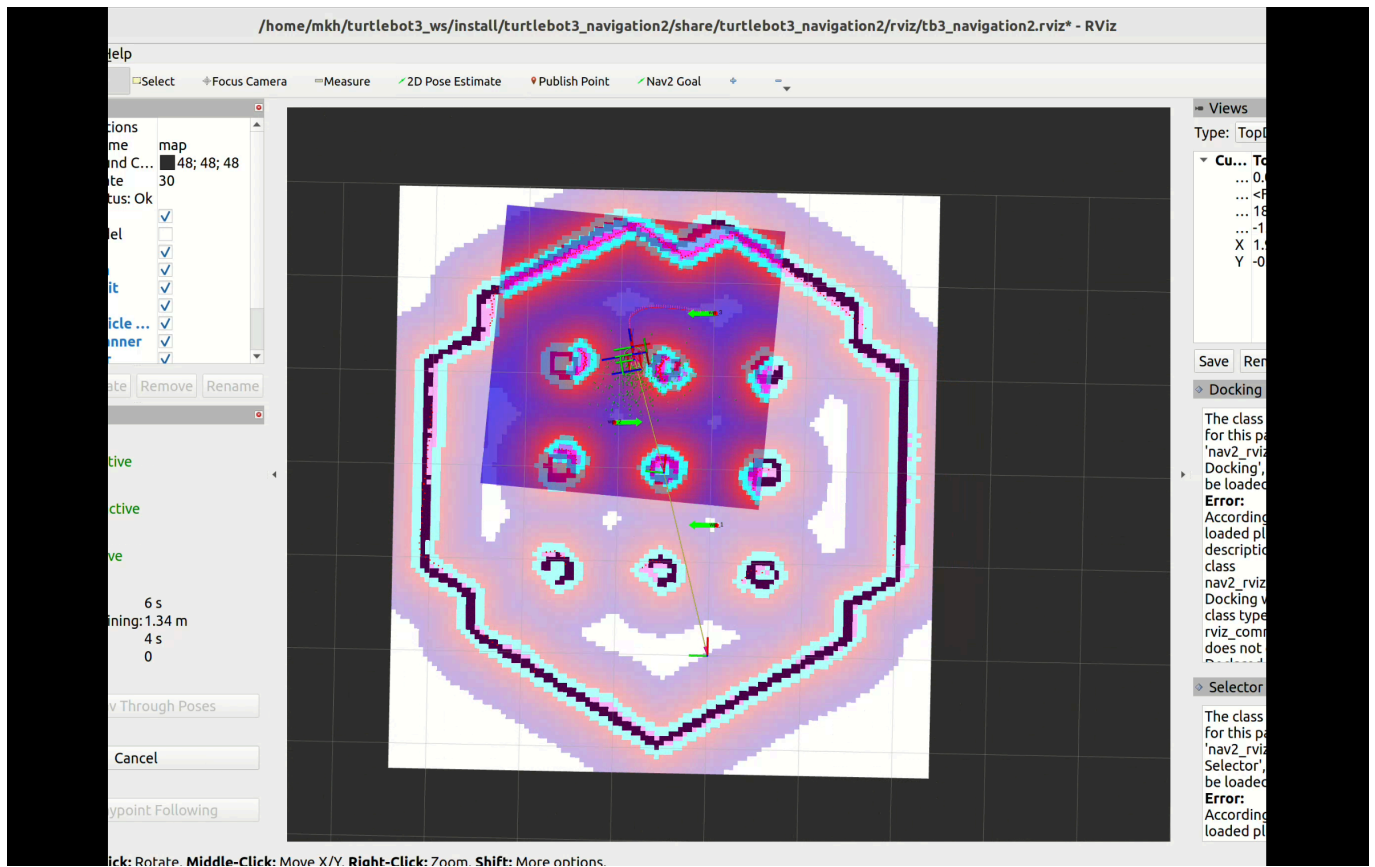


그리고 아래 버튼을 누르면



좌표를 지정한 순서대로 주행하기 시작한다

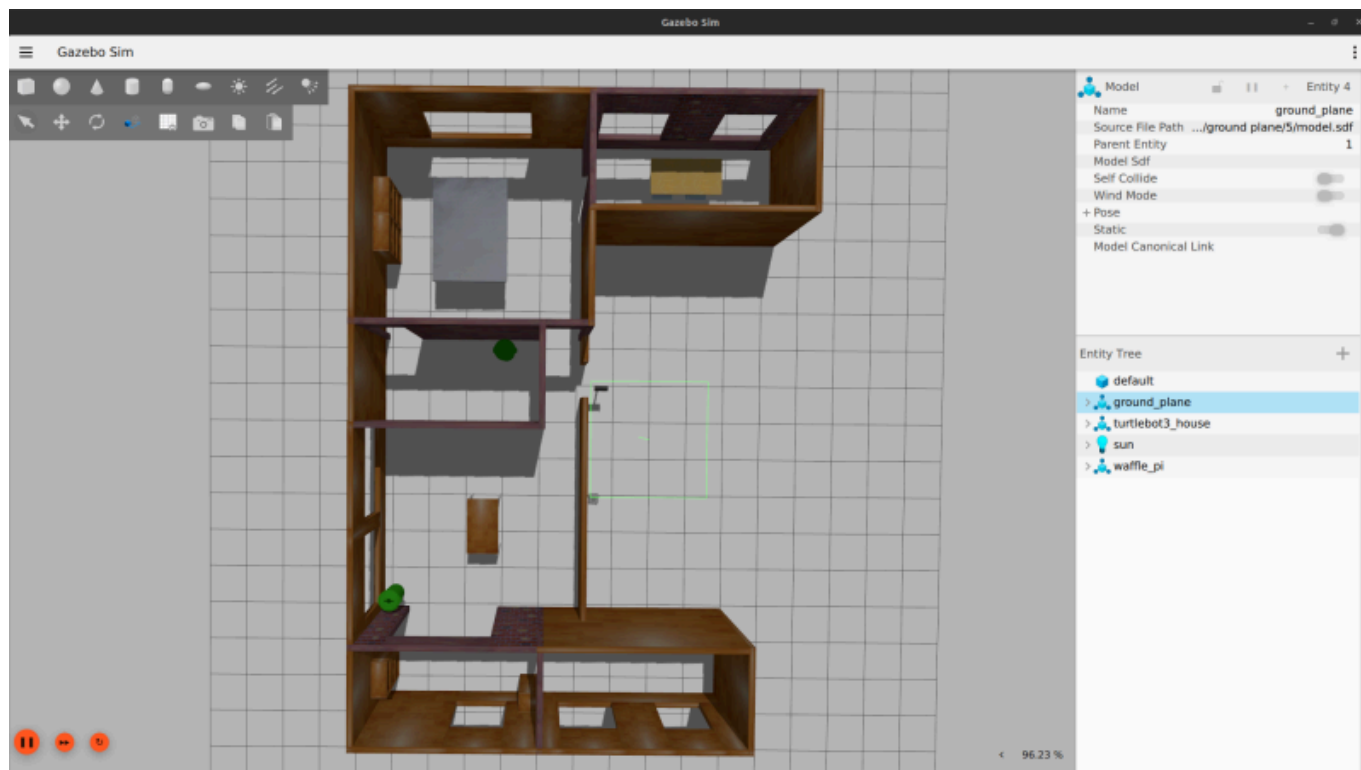




다양한 월드에서 연습해보자~

Code

```
1 ros2 launch turtlebot3_gazebo turtlebot3_house.launch.py
```





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